Requirements Gathering

Customer Churn Analysis and Retention Strategy for the Olist E-Commerce Marketplace

Project title	Customer Churn Analysis and Retention Strategy (Olist Brazilian E-Commerce dataset)
Team name	Customer Churn
Track	Data Engineering
Team Leader	Ziad Khaled Abd El-Rahman
Team members (4 to 6)	1. Ziad Khaled Abd El-Rahman 2. Omar Hassan Ibrahim 3. Amr Mostafa Etman 4. Malak Ashraf Mostafa 5. Mennatullah Hesham Mohamed 6. Ahmed Ayman Naguib
Lecturer / Instructor	Ahmed Essam Azab
GitHub repository	https://github.com/nhahub/NHA-5-086
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1. Introduction

1.1 Purpose

This document lists what the churn analysis and dashboard must do and the qualities it must have. It turns the proposal and plan in the Project Planning & Management document into stakeholders, user stories, use cases, and functional and non-functional requirements. It is the base for the Design document (due 6 Nov 2026) and for the test cases.

1.2 Scope in brief

The team will clean and combine the 9 Olist tables, create a churn column with the 90-day rule, explore what is linked to churn, build a simple interactive dashboard (Power BI or Streamlit), and write a retention strategy report based on the findings. A simple prediction model is optional. Real-time data, production deployment, a mobile app, paid tools, real campaigns or A/B tests, and data from outside Olist are out of scope.

1.3 Definitions

Term	Meaning
Churn	A customer who does not place another order within 90 days after their first delivered order (churn = 1).
T0	The date of the customer's first delivered order.
Customer identifier	customer_unique_id from the customers table. The customer_id column changes with every order.
Observation window	The 90 days after T0. Customers whose window cannot be fully observed are removed.
Class imbalance	The situation where one class is much larger than the other. Here most customers have churn = 1.
KPI	Key Performance Indicator, a number used to judge the project or the business.
MoSCoW	Priority scale: Must have, Should have, Could have.

2. How the Requirements Were Gathered

Olist is a public dataset, so the team does not have a real client to interview. The business users (S3 to S5 below) are therefore represented by roles that a company like Olist would have, and their needs are taken from the sources below. This is stated clearly so that the lecturer can judge it during the review.

Source or technique	What it gave us	Status
Planning document (proposal, scope, objectives, KPIs, risks)	The scope, the churn rule, the deliverables and the targets used in the requirements	Done
Literature review	The variables linked to repeat purchase, the class-imbalance and data-leakage issues, and dashboard design advice	Done
Dataset documentation and exploration of the 9 tables	The data available for each requirement, for example order status, delivery dates, review score and payment type	Done
Role-based users and scenarios	The user stories and use cases in Sections 4 and 5	Done
Lecturer review	Confirmation of the churn rule, the scope and the requirements	Planned, 2 Nov 2026
Test with at least 3 users	Check that the dashboard can be used with the manual only	Planned, testing phase

3. Stakeholders

3.1 Stakeholder list

ID	Stakeholder	Type	Interest	Main needs from the project
S1	Lecturer / Instructor	Evaluator	Clear, defensible churn label; complete and on-time documents; every member's work visible	Approves requirements; checks the 90-day rule
S2	Project team (6 members)	Builder	A realistic scope that can be finished by 4 Dec 2026; clear roles	Build, test and document the solution
S3	Retention / Marketing Manager (business user, represents Olist)	Primary user	Find which customers leave and why; decide where to spend retention money	KPI cards, churn by segment, filters, retention actions
S4	Operations / Logistics Manager (represents Olist)	Primary user	Know whether late delivery and low review scores push customers away	Churn by delivery status and delay; churn by review score
S5	Business Owner / Executive (represents Olist)	Secondary user	A short, honest view of the problem and the expected benefit of actions	Highest-churn segments, class-imbalance report, expected benefit
S6	Program evaluators (final presentation)	Evaluator	Complete deliverables, working demo, each member explains own part	Working dashboard, documents, GitHub repository
S7	Test users (at least 3)	Tester	A dashboard that can be used without help	User manual; simple layout

3.2 User roles used in the stories

Role	Description
Retention Manager	Plans offers and campaigns to keep customers. Is comfortable with charts but not with code. Wants to know which groups leave and where to act.
Logistics Manager	Responsible for delivery. Wants to know if delivery delay and low review scores affect repeat purchase.
Business Owner	Wants a short summary, the biggest problem areas and the expected benefit. Has little time.

Role	Description
Team roles	Ziad Khaled Abd El-Rahman: Team Leader and Data Engineer Omar Hassan Ibrahim: Data Analyst Amr Mostafa Etman: Data Scientist Malak Ashraf Mostafa: Dashboard Developer Mennatullah Hesham Mohamed: Testing & QA Ahmed Ayman Naguib: Documentation Lead Roles are defined in Section 3 of the planning document.

4. User Stories

Priority follows the MoSCoW scale. Must items are needed for the project to be accepted. Should items are important but the project can still work without them. Could items are done only if time allows.

ID	User story	Priority	Acceptance criteria
US-01	As a **Retention Manager**, I want to see the overall churn rate and repeat purchase rate in KPI cards, so that I understand the size of the problem at a glance.	Must	Cards show values that match the cleaned dataset; the churn definition is visible on the page.
US-02	As a **Retention Manager**, I want to filter the dashboard by state, product category, payment type, review score, delivery status and order date, so that I can focus on one customer group.	Must	Every chart and KPI updates after a filter is changed; response time is under 3 seconds.
US-03	As a **Retention Manager**, I want to see churn rate by customer state, so that I can find regions with a high loss of customers.	Must	A chart lists states with their churn rate and number of customers; values match QA's check.
US-04	As a **Retention Manager**, I want to see churn rate by product category, so that I know which first purchases lead to repeat buying.	Must	A chart shows the main categories with churn rate and customer count; small categories are grouped.
US-05	As a **Logistics Manager**, I want to compare churn for on-time and late deliveries, so that I can judge whether delivery delay pushes customers away.	Must	The dashboard shows churn rate for on-time vs late orders and by delay range.
US-06	As a **Logistics Manager**, I want to see churn rate by review score, so that I can see the effect of customer satisfaction.	Must	A chart shows churn rate for each score from 1 to 5.
US-07	As a **Retention Manager**, I want to see churn by payment type and installments, so that I can check whether payment options matter.	Must	A chart shows churn rate by payment type and by number of installments.
US-08	As a **Business Owner**, I want to see the highest-churn segments with their size, so that I can prioritize where to act.	Must	A table ranks segments by churn rate and shows the number and share of customers in each.
US-09	As a **Business Owner**, I want a class-imbalance report, so that I understand why most customers show as churned and how to read the numbers.	Must	The dashboard and the report show the share of churn = 1 and churn = 0, with a short explanation.
US-10	As a **Retention Manager**, I want a retention strategy report where each action is linked to a dashboard finding, so that I can trust and use the advice.	Must	Each action names the finding that supports it and the expected benefit; at least 5 insights are included.
US-11	As the **Lecturer**, I want the churn rule and all cleaning rules documented, so that I can check that the label is correct.	Must	The rule in Section 1.4 of the planning document is repeated in the repository with every cleaning rule listed.
US-12	As the **Data Engineer**, I want a repeatable process that loads the 9 tables and creates the churn column, so that the result is the same each time it is run.	Must	Running the process on the raw files gives the same cleaned file; repeat customers are found by customer_unique_id.
US-13	As **Testing & QA**, I want test cases for data, metrics and the dashboard, so that errors are found before the presentation.	Must	Test report exists; at least 90% of test cases pass; bugs are logged and re-checked.

ID	User story	Priority	Acceptance criteria
US-14	As a **Test User**, I want a short user manual, so that I can answer questions with the dashboard without help.	Should	At least 3 test users find the answers to given questions using only the manual.
US-15	As the **Lecturer**, I want to see how the result changes with 60 and 180 days, so that I can judge how sensitive the 90-day rule is.	Could	A table or chart compares churn rate for 60, 90 and 180 days.
US-16	As a **Data Scientist**, I want a simple prediction model with proper metrics, so that I can show which customers are at risk.	Could	If built, Accuracy, Precision, Recall, F1-score and ROC-AUC are reported; class weights are used.

5. Use Cases

5.1 Actors and use case list

ID	Use case	Main actor	Related stories
UC-01	View churn overview	Retention Manager, Business Owner	US-01, US-09
UC-02	Filter and explore churn by segment	Retention Manager	US-02, US-03, US-04, US-07, US-08
UC-03	Compare delivery and review results with churn	Logistics Manager	US-05, US-06
UC-04	Review the class-imbalance report	Business Owner	US-09
UC-05	Read the retention recommendations	Retention Manager, Business Owner	US-10
UC-06	Prepare the churn dataset	Data Engineer (Ziad Khaled Abd El-Rahman)	US-11, US-12
UC-07	Run tests and log bugs	Testing & QA (Mennatullah Hesham Mohamed)	US-13

UC-01, UC-02, UC-03, UC-05 and UC-06 are described in detail below. UC-04 and UC-07 are simple and are described through their user stories and requirements.

UC-01 View churn overview

Actor	Retention Manager, Business Owner
Goal	See how many customers churn and how many come back.
Precondition	The cleaned dataset with the churn column is loaded in the dashboard.
Main flow	1. The user opens the dashboard. 2. The system shows the KPI cards: churn rate, repeat purchase rate, average order value and average review score. 3. The system shows the class balance and the note about the 90-day rule. 4. The user reads the numbers.
Alternative flow	If the data cannot be loaded, the dashboard shows a clear message instead of empty charts, and the team uses the backup screenshots.
Postcondition	The user understands the size of the churn problem.
Requirements	FR-11, FR-17, FR-20, FR-23, NFR-01

UC-02 Filter and explore churn by segment

Actor	Retention Manager
Goal	Find the customer groups with the highest churn.

Precondition	The dashboard is open (UC-01).
Main flow	1. The user selects one or more filters: state, category, payment type, review score, delivery status or order date. 2. The system updates all KPI cards and charts to the selected group. 3. The user reads churn rate by state, category and payment type. 4. The user opens the table of highest-churn segments and reads their size and share.
Alternative flow	If the selected filters match no customers, the system shows "No customers match these filters" and keeps the filters so the user can change them.
Postcondition	The user knows which segments to target.
Requirements	FR-13, FR-18, FR-19, FR-21, NFR-02, NFR-04

UC-03 Compare delivery and review results with churn

Actor	Logistics Manager
Goal	Check whether late delivery and low review scores are linked to churn.
Precondition	The dashboard is open and delivery delay and review score are available for the customers.
Main flow	1. The user opens the delivery section. 2. The system shows churn rate for on-time and late deliveries and by delay range. 3. The user opens the review section and sees churn rate for each score from 1 to 5. 4. The user combines both with a filter to compare groups.
Alternative flow	Orders without a review are shown in their own group so that they are not hidden.
Postcondition	The user can state whether delivery or satisfaction is linked to churn, using numbers.
Requirements	FR-14, FR-18, FR-19

UC-05 Read the retention recommendations

Actor	Retention Manager, Business Owner
Goal	Get clear actions that are backed by the data.
Precondition	The dashboard and the retention strategy report are finished.
Main flow	1. The user opens the retention strategy report. 2. The report lists each action with the dashboard finding that supports it. 3. The report shows the expected benefit of each action. 4. The user checks the finding on the dashboard.
Alternative flow	If a finding is small or unclear, the report says so instead of overstating it.
Postcondition	The user can choose which actions to try first.
Requirements	FR-15, FR-22, FR-26

UC-06 Prepare the churn dataset

Actor	Data Engineer: Ziad Khaled Abd El-Rahman (backup: Amr Mostafa Etman, Data Scientist)
Goal	Produce one clean customer-level table with the churn column.
Precondition	The 9 Olist files are downloaded from Kaggle.
Main flow	1. The Data Engineer loads the 9 tables and links them. 2. The system keeps delivered orders only and uses customer_unique_id. 3. The system finds each customer's first delivered order (T0). 4. The system removes customers with less than 90 days of observation. 5. The system creates churn = 1 or 0 with the 90-day rule. 6. The Data Engineer cleans missing values and duplicates and records every rule.

	7. The system saves the customer-level table to GitHub.
Alternative flow	If the data test finds no repeat customers, the process stops and the customer identifier is checked before going on.
Postcondition	The cleaned dataset and the cleaning rules are available to the whole team (target date 13 Nov 2026).
Requirements	FR-01 to FR-10, NFR-03, NFR-08

6. Functional Requirements

The column "Source" shows where in the planning document the requirement comes from. Requirements marked "Team analysis" were added by the team to support a user story.

6.1 Data preparation

ID	Requirement	Priority	Source
FR-01	The system shall load all 9 Olist tables (orders, order items, payments, reviews, customers, sellers, products, geolocation, product category name translation) from the Kaggle files.	Must	Plan 1.3, KPI 5.1
FR-02	The system shall identify customers by customer_unique_id, not customer_id.	Must	Plan 1.4, Risk 2
FR-03	The system shall count only orders with status delivered and exclude canceled and unavailable orders.	Must	Plan 1.4
FR-04	The system shall use order_purchase_timestamp as the order date and set T0 as the date of the customer's first delivered order.	Must	Plan 1.4
FR-05	The system shall create the column churn: 1 if the customer places no further order within 90 days after T0, and 0 if a second order is placed within 90 days after T0.	Must	Plan 1.4
FR-06	The system shall remove customers whose first order is less than 90 days before the last order date in the dataset.	Must	Plan 1.4, Risk 3
FR-07	The system shall handle missing values, duplicates and inconsistent values, and every cleaning rule shall be recorded.	Must	Plan 1.5, Risk 7
FR-08	The system shall produce one customer-level table (one row per customer) with the churn column and first-order information: order value, freight cost, number of items, payment type, installments, product category, customer state, delivery delay, late-delivery flag and review score.	Must	Plan 1.4, 1.5
FR-09	The system shall keep only the columns it needs and use aggregated tables, so that large files such as geolocation do not slow the tools.	Should	Plan Risk 8
FR-10	The system shall include a data test that confirms repeat customers are found after the customer identifier is applied.	Must	Plan Risk 2, KPI 5.2

6.2 Exploration and analysis

ID	Requirement	Priority	Source
FR-11	The system shall calculate the overall churn rate, the repeat purchase rate and the class balance.	Must	Plan 1.4, 5.4
FR-12	The team shall produce a class-imbalance report.	Must	Plan 1.5, KPI 5.2
FR-13	The system shall calculate churn rate for at least 5 breakdowns: state, product category, payment type, review score and delivery status.	Must	Plan KPI 5.2
FR-14	All breakdowns shall use only information known from the first order.	Must	Plan 1.4, Risk 4
FR-15	The team shall write at least 5 data-backed insights about customer behavior.	Must	Plan 1.7, KPI 5.2

ID	Requirement	Priority	Source
FR-16	The team shall deliver an EDA notebook with visualizations (bar charts, pie charts, histograms) and a summary report of key insights.	Must	Plan 1.7 (Milestone 1)

6.3 Dashboard

ID	Requirement	Priority	Source
FR-17	The dashboard shall show KPI cards for overall churn rate, repeat purchase rate, average order value and average review score.	Must	Plan 1.5, 5.4
FR-18	The dashboard shall show charts of churn rate by state, product category, payment type and review score, and for on-time versus late deliveries.	Must	Plan 5.4
FR-19	The dashboard shall provide filters (state, category, payment type, review score, delivery status, order date) and all charts shall update when a filter changes.	Must	Plan 1.5, KPI 5.3
FR-20	The dashboard shall show the class balance with a short note that explains why churn is high.	Must	Plan 1.4, KPI 5.2
FR-21	The dashboard shall list the highest-churn segments with the number and share of customers in each.	Must	Plan 5.4
FR-22	The dashboard shall show the expected benefit of the proposed retention actions.	Should	Plan 5.4
FR-23	The dashboard shall display the churn definition (90-day rule) in a visible note or help text.	Should	Team analysis (supports US-01, US-11)
FR-24	The dashboard shall be built with one free tool chosen by the team: Power BI Desktop or Streamlit.	Must	Plan 2.4
FR-25	The team shall keep dashboard screenshots and a recorded demo as a backup for the presentation.	Should	Plan Risk 9

6.4 Reporting and documentation

ID	Requirement	Priority	Source
FR-26	The team shall write a retention strategy report in which each action is linked to a dashboard finding.	Must	Plan 1.5, 1.7
FR-27	The team shall keep the source code, data files and documents in a GitHub repository with a README.	Must	Plan 1.7, 3
FR-28	The team shall write test cases (data, metric and dashboard checks), run them, and deliver a test report and a bug report.	Must	Plan 1.7, 2.1
FR-29	The team shall write a user manual.	Must	Plan 1.7, KPI 5.3
FR-30	The team shall deliver a final presentation in which each member explains the work they completed.	Must	Plan 1.7, 3

6.5 Optional features

ID	Requirement	Priority	Source
FR-31	The team may repeat the churn labeling with 60 and 180 days and compare the results with the 90-day label.	Could	Plan 1.4, 1.6
FR-32	The team may build a simple churn prediction model (for example Logistic Regression) with class weights and report Accuracy, Precision, Recall, F1-score and ROC-AUC.	Could	Plan 1.6, Risk 1, KPI 5.2

7. Non-Functional Requirements

ID	Category	Requirement	Target and source
NFR-01	Performance	The dashboard page shall load in under 5 seconds.	Measured on the presentation machine; Plan KPI 5.3
NFR-02	Performance	The dashboard shall respond to a filter change in under 3 seconds.	Measured in test runs; Plan KPI 5.3
NFR-03	Data quality	All 9 tables shall be loaded and linked, and key fields shall have less than 1% unresolved missing values.	100% of tables; under 1% missing; Plan KPI 5.1
NFR-04	Correctness	The churn label and all dashboard figures shall be correct, and filters shall update all charts.	Label verified by QA; filters correct in all test runs; Plan KPI 5.2, 5.3
NFR-05	Usability	A user shall be able to answer business questions with the dashboard using only the user manual. KPI cards shall be the first thing the user sees.	At least 3 test users succeed; Plan KPI 5.3
NFR-06	Reliability	The presentation shall not depend on one live setup. The dashboard shall be tested early on the machine used for the presentation.	Screenshots and recorded demo available; Plan Risk 9
NFR-07	Maintainability	The code shall be stored in GitHub with feature branches, pull requests and clear commit messages, and the README shall explain how to run the project.	All members have documented tasks and commits; Plan KPI 5.1
NFR-08	Reproducibility	Running the data preparation again on the raw files shall give the same cleaned dataset and the same churn rates.	Same results in two runs; team requirement
NFR-09	Compatibility	The solution shall run on the team's own laptops using free tools only and shall not need a server or cloud service.	Works on the presentation machine; Plan 2.4
NFR-10	Security and privacy	Only the public, anonymized Olist dataset shall be used. No personal data shall be collected, and no passwords or keys shall be stored in the repository.	Checked in the final review; Plan 1.6
NFR-11	Testability	Each Must requirement shall have at least one test case.	At least 90% of test cases pass; Plan KPI 5.1
NFR-12	Cost	No paid tools or services shall be used.	Budget of zero; Plan 2.4
NFR-13	Schedule	Each deliverable shall be submitted by its planned date.	100% on time; Plan KPI 5.1

8. Constraints, Assumptions and Dependencies

8.1 Constraints

- Only the Olist dataset is used. No data from outside the dataset.
- Only free and open-source tools are used. The budget is zero.
- All deliverables are due by 4 Dec 2026, with Requirements Gathering planned for 4 Nov 2026.
- Real-time or live data is out of scope.

8.2 Assumptions

- The Olist dataset is used as downloaded from Kaggle, and the team has free access to it.
- The 90-day churn rule is accepted by the lecturer or adjusted in the lecturer review.
- All members are available for about 6 to 8 hours per week until 4 Dec 2026.
- Most customers will have churn = 1, so the class balance is shown clearly.

8.3 Dependencies

- Design (ERD, architecture, wireframes) depends on the approved requirements.
- The dashboard depends on the cleaned dataset (target 13 Nov 2026).
- The retention strategy report depends on the finished dashboard findings.
- Test cases depend on the acceptance criteria in Section 4.

9. Traceability

The table connects each project objective in Section 1.5 of the planning document to the stories, requirements and checks that cover it.

Objective	User stories	Requirements	How it is checked
Clean and combine the 9 tables into one customer-level data model	US-12	FR-01 to FR-04, FR-07 to FR-10	Data tests; KPI 5.1
Create the churn column with the 90-day rule and document it	US-11, US-12	FR-05, FR-06, FR-10, FR-31	QA label check; KPI 5.2
Explore the data and produce the class-imbalance report	US-09	FR-11 to FR-16	Review of EDA notebook and report
Build the interactive dashboard	US-01 to US-08	FR-17 to FR-25, NFR-01, NFR-02, NFR-04, NFR-05	Dashboard test cases; KPI 5.3
Write the retention strategy report	US-10	FR-15, FR-22, FR-26	Review by Documentation Lead and lecturer
Deliver documentation, testing, user manual and final presentation on time	US-13, US-14	FR-27 to FR-30, NFR-07, NFR-11, NFR-13	Test report; deadline tracking

10. Priority Summary

Item	Must	Should	Could	Total
User stories	13	1	2	16
Functional requirements	26	4	2	32
Non-functional requirements	13	0	0	13

11. Open Points for the Lecturer Review

- **Churn rule.** Confirm the 90-day rule, or give the window the lecturer prefers.
- **Stakeholders.** Confirm that role-based users are acceptable, since Olist is a public dataset and there is no real client to interview.
- **Dashboard tool.** The team must choose Power BI Desktop or Streamlit. Power BI Desktop runs only on Windows, so the choice depends on the laptops used for the presentation.
- **Optional items.** Confirm whether the simple prediction model (FR-32) and the 60/180-day check (FR-31) are wanted or should be dropped to protect the deadline.
- **Team details.** Confirm that the team name, track, lecturer name and GitHub link on the first page are correct.

12. Approval

Prepared by (Team Leader)	Ziad Khaled Abd El-Rahman Signature: _____ Date: 31 Oct 2026
Reviewed by (Lecturer)	Ahmed Essam Azab Signature: _____ Date: _____